ERPs System Documentation

miacorp

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1. Overview

Following Earth's collapse and the subsequent human migration to Mars, MiaCorp has become the cornerstone of Martian civilization. With responsibility spanning from resource allocation to population management and terraformation research, the company must ensure survival through intelligent data governance and sustainable planning. This document outlines the architecture, usage, and functional modules of the MiaCorp ERPs System, developed to manage the critical needs of the Martian colony. More than just software, this ERPs is a centralized lifeline — integrating civil registration, supply tracking, human resource planning, and AI-assisted decision-making to maintain social order and biological viability in a fragile extraterrestrial environment.

2. System Architecture

- **Frontend:** React.js with custom JavaScript for low-latency operations (custom UI/UX tailored for Mars-based low-latency operations).
- **Backend:** Low-code Retool + serverless API layers.
- **Database:** MySQL.
- **Deployment:** Google Cloud Platform, Starlink network-compatible.
- **Security:** Humanity Agile Control Protocol (HACP).

3. User Roles & Access

To maintain operational integrity and data security in MiaCorp's Martian ERP system, access is governed by clearly defined user roles. Each role is associated with specific modules and permissions, ensuring users interact only with data and features relevant to their responsibilities.

Role	Module Access	Permissions	Description
Administrator	All modules (CRM, SCM, FIN, HR)	Full Create/Read/Update/Delete (CRUD); System configuration	Oversees the entire system, performs data audits, manages user accounts, and enforces global policies.

Mars Coordinator	CRM	Register citizens, manage resource requests, trigger AI approvals	First contact for new arrivals; handles onboarding, daily quota approvals, and family planning applications.
Procurement Officer	SCM	Supplier registry, stock tracking, procurement logs	Ensures continuous availability of essential supplies like oxygen tanks, food packets, and water filtration units.
Finance Analyst	FIN	Ledger entries, financial visualization, cost center review	Manages interdepartmental budgets, expenses, and sustainability forecasting.
HR Director	HR	Applicant tracking, department trials, AI promotion review	Oversees all HR operations, configures AI rules for job matching, and monitors the colony's human capital pool.
Department Head	HR (Limited)	Trial feedback, performance reviews, promotion suggestions	Evaluates trial-phase interns or staff and helps identify future leaders for colony expansion roles.
Hiring Manager	HR (Limited)	Review candidates, conduct interviews, update status	Focused on department-specific recruitment and competency evaluation.
Martian Citizen	CRM (Self-Service)	View profile, submit requests, check quota balance	Access to personal dashboard for managing requests

			(e.g., leave, housing, family planning).

3.1 Access Management Features:

- **Role-Based Access Control (RBAC):** Ensures data privacy and operational separation of concerns.
- **Audit Logging:** All high-level actions are time stamped and traceable for accountability.
- **AI-Guarded Interfaces:** Sensitive approvals (e.g., childbirth or resource override) require AI recommendation and human validation.
- **Emergency Override Protocol:** In catastrophic scenarios, select Admins can activate override to reassign roles and permissions.

4. Module Functions

Module	Purpose
CRM	Citizen onboarding, request handling (housing, medical, family planning), and individual resource monitoring
SCM	Martian inventory control, procurement from Earth or domes, and supplier traceability
FIN	Financial integrity through ledgers, budget planning, and inter-departmental cost tracking
HR	Talent acquisition, departmental trials, AI job matching, and long-term growth planning for repopulation

4.1 Customer Relationship Management (CRM) Module

4.1.1 Module Purpose:

The CRM module in MiaCorp's ERP system is designed to manage Mars citizen data, resource requests, and quota tracking. It supports Mars coordinators in handling citizen onboarding, service requests, and sustainability management. This module forms the basis for tracking human integration and survival metrics on Mars

4.1.2 Module Functions:

4.1.2.1 Core CRM Capabilities

- **Citizen Registration:** Store and manage core demographic data (e.g., name, gender, origin, status).
- **Request Management:** Log and monitor individual requests such as housing, medical needs, or family planning.
- **Lead Contact Tracking:** Record first contact information from new arrivals or referrals.
- **Resource Monitoring:** Track and manage each citizen's daily water, oxygen, and food allocations.

4.1.2.2 AI Function: Request Approval Recommendation

- **Purpose**: Help coordinators decide whether to approve requests based on resource availability.
- **Implementation**: In our CRM, two buttons (Family Planning / Housing Allocation) trigger AI queries. When pressed, the system fetches citizens under "Under Review" status for the selected request type. AI automatically generates a recommendation on whether to approve or defer each request — based on priority, water, oxygen, and food thresholds.

4.1.2.3 Data Visualization: Resource Quota Usage

- **Purpose**: Enable coordinators to monitor sustainability by visualizing per-citizen resource usage.
- **Implementation**: A bar chart displays water and oxygen consumption of each citizen, helping decision-makers evaluate high-usage individuals and plan better distribution.

4.1.3 Entities and Attributes:

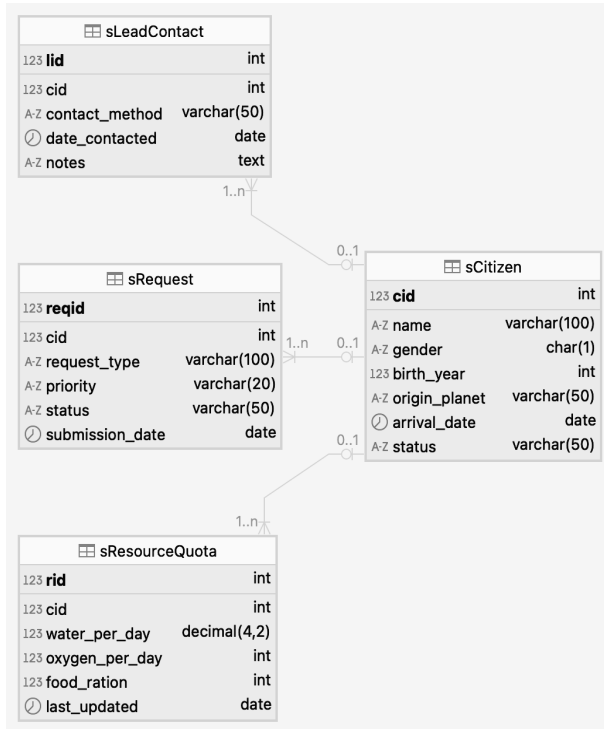
Citizen:
Stores demographic and origin information of each Mars resident, including their ID, name, gender, birth year, origin planet, arrival date, and status. This forms the foundation of all citizen-related records.

Request:
Tracks resource and service-related requests submitted by citizens. It records the type, urgency, status, and submission date of each request, and links each one to a citizen using cid.

Lead Contact:
Log initial communication details when a citizen first makes contact with MiaCorp. It includes contact method, date, and any interview notes, helping the CRM trace the onboarding process.

ResourceQuota:
Monitors and updates each citizen's daily allocation of essential resources — water, oxygen, and food. It is used to evaluate sustainability and determine eligibility for certain requests.

ER diagram:



4.2 Supply Chain Management (SCM) Module

4.2.1 Module Purpose:

The Supply Chain Management (SCM) module facilitates the streamlined coordination of inventory, procurement, and supplier data to ensure products are sourced, stocked, and delivered efficiently. It acts as the backbone of the product fulfillment cycle, connecting suppliers, warehouse operations, and buyers via an integrated order system.

4.2.2 Core Functionalities:

4.2.2.1 Inventory Management

- Monitors real-time stock levels.
- Updates stock quantities automatically based on order fulfillment.
- Tracks product categories and selling prices for reporting and analytics.

4.2.2.2 Procurement

- Ensures products are consistently sourced from predefined suppliers.
- Captures procurement metadata for each product (e.g., supplier ID, product cost).
- Prevents multi-supplier conflicts by enforcing a one-product-one-supplier rule.

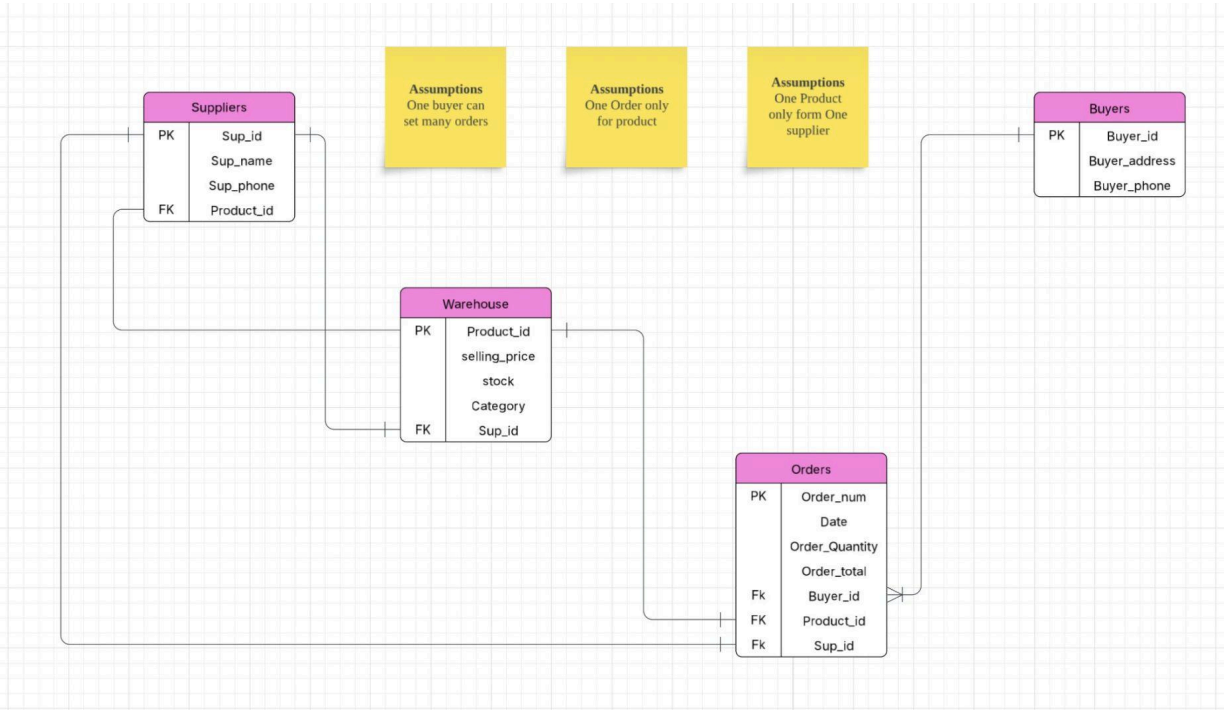
4.2.2.3 Supplier Management

- Maintains comprehensive supplier contact and performance data.
- Links each supplier directly to the products they provide.
- Enables traceability of supply sources and supports negotiation leverage.

4.2.2.4 Order Fulfillment

- Handles buyer orders, quantity validation, and stock deductions.
- Computes order totals based on quantity and current selling price.
- Ensures consistency by validating supplier and product mappings per order.

4.2.3 Physical ER diagram:



4.2.4 Data Model and Key Entities:

- **Product description:** This table contains detailed information about each product, including its category, total stock, and supplier ID. It is linked to the Orders table through the product ID (as the primary key) and the supplier ID (as a foreign key).
- **Orders:** This table records all order activities, including the order date, quantity, and buyer ID. It is connected to the Product Description table via the product ID and supplier ID to ensure accurate tracking of each transaction.

4.2.5 Entity Relationships:

Entity	Description
Suppliers	Each supplier provides specific products. Enforced one-to-one mapping prevents multi-vendor duplication for the same product.
Warehouse	Centralized storage that records current stock, pricing, and categories. Linked to a single supplier per product.
Orders	Operational record of all purchases made by buyers. Each record is tied to one buyer, one product, and its designated supplier.

- **Buyer ↔ Orders:** One buyer can place multiple orders. Enables tracking of customer purchasing history and behavior.
- **Orders ↔ Product ↔ Suppliers:** Every order references a product tied to a single supplier. This structure ensures data integrity and simplifies procurement tracking.
- **Warehouse ↔ Suppliers:** Each product in the warehouse is tied to one supplier, reflecting a real-world scenario where vendors are product-specific.

4.2.6 CRUD:

4.2.6.1 CRUD in Our Order & Warehouse System

Our system uses full CRUD functionality (Create, Read, Update, Delete) to manage and connect the Order Table and Warehouse Table seamlessly.

4.2.6.2 Order Table (Customer Orders)

CRUD Function	Description
Create	Users can place a new order using a form. The form allows for user inputs for product id, order date, buyer id, and the quantity for the product. After the user hits submit, a query activates which gets the supplier id based on the product id, it automatically assigns an order number by incrementing the previous order number by 1, and it calculates the total cost of the order by multiplying the product selling price with the order quantity. Lastly, it also automatically updates the orders table.
Read	Users can view and filter past orders by product, buyer, or date using table features.
Update	If there's an error or a change, users can edit order details directly within the table.
Delete	Unwanted or incorrect orders can be selected and deleted from the system.

4.2.6.3 Warehouse Table (Inventory Tracking)

CRUD Function	Description
Create	New product entries, along with their stock, supplier, and category, can be added to the Warehouse Table.
Read	Users can search or filter product listings to check stock levels and supplier details.
Update	When an order is placed, the stock is automatically updated. Users can also manually update fields like supplier or stock situation.
Delete	Obsolete or discontinued products can be removed from the table.

4.2.6.4 Linking Between Tables

- When a new order is created, the Order Quantity is automatically deducted from the Warehouse Stock of the matching Product_ID.
- This keeps inventory accurate and reduces the chance of overselling or stockouts.

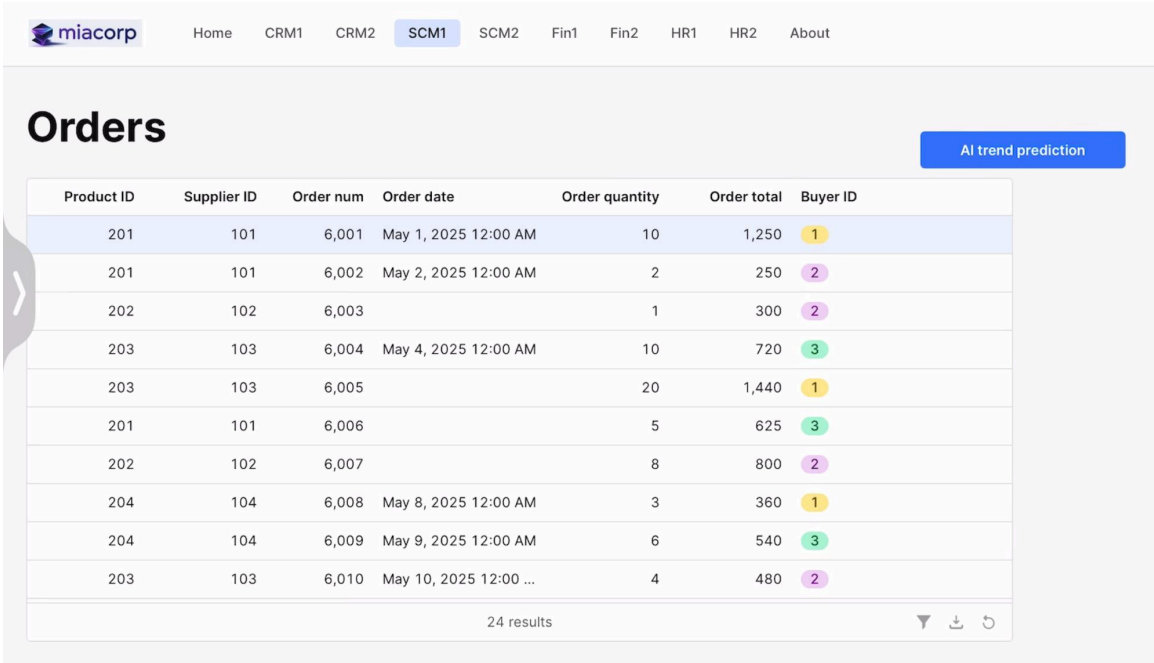
4.2.7 AI:

As part of the AI-powered supply chain system, the AI model analyzes current trends using data from the CRUD system. This includes information such as product ID, order quantity, and order timestamps.

By observing these patterns, the AI can detect trends such as:

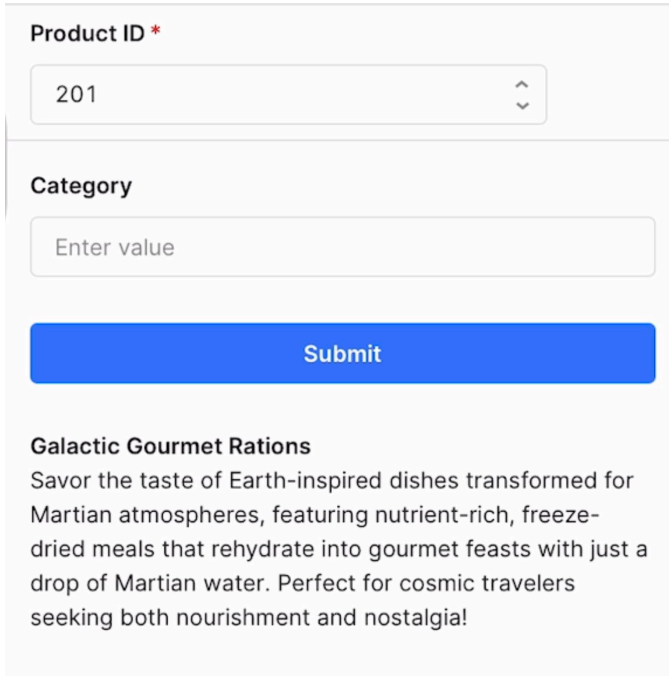
- Increased order frequency.
- Periodic spikes in demand (e.g., monthly or seasonal).
- Typical order volume ranges.

These insights allow the system to predict future demand trends, enabling better inventory planning, procurement decisions, and resource allocation. The AI system in SCM enhances procurement planning and product classification by identifying order patterns and product correlations.



Product ID	Supplier ID	Order num	Order date	Order quantity	Order total	Buyer ID
201	101	6,001	May 1, 2025 12:00 AM	10	1,250	1
201	101	6,002	May 2, 2025 12:00 AM	2	250	2
202	102	6,003		1	300	2
203	103	6,004	May 4, 2025 12:00 AM	10	720	3
203	103	6,005		20	1,440	1
201	101	6,006		5	625	3
202	102	6,007		8	800	2
204	104	6,008	May 8, 2025 12:00 AM	3	360	1
204	104	6,009	May 9, 2025 12:00 AM	6	540	3
203	103	6,010	May 10, 2025 12:00 ...	4	480	2

2. On top of that, to allow the user to easily find what kind of product is due to the fact that we would have many products in our inventory in the future, it may somehow lead to confusion. Also, users may forget which product ID is related to which product line. Therefore, AI can allow us to help the user understand what each product is and how it works in order for smoother operations.



4.3 Finance (FIN) Module

4.3.1 Module Purpose:

The Finance (FIN) module is designed to support the accounting and financial operations within MiaCorp's ERP system. It enables finance personnel to manage ledger entries, monitor cost centers, and maintain overall financial integrity through structured input and smart validation. It is also crucial for financial accounting and reporting to the interplanetary tax authorities and company owners.

4.3.2 Module Functions:

4.3.2.1 General Ledger Management

- Allows finance users to manually input journal entries including account, amount, date, and cost center, etc.
- Supports classification of debit and credit entries to maintain accurate financial records.
- Ensures consistent formatting and structure through a guided form interface.

4.3.2.2 Debit and Credit Visualization

- Displays a live-updating pie chart to show the current ratio between debit and credit transactions.
- Reflects real-time data from the General Ledger to support ongoing financial monitoring.

4.3.2.3 Cost Center Activity Tracking

- Aggregates and displays the number of transactions per cost center in a bar chart format.
- Highlights the most active departments based on financial transaction volume.
- Assists finance teams in evaluating departmental resource usage and budgeting priorities.
- Manages organizational units with budgets and planetary locations.

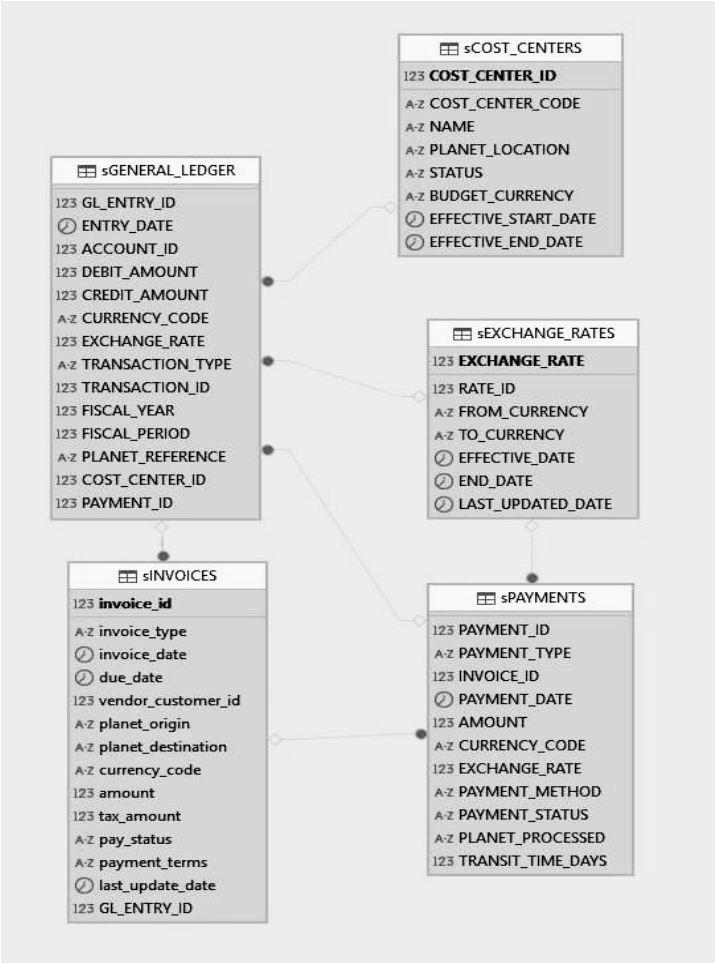
4.3.2.4 Currency Exchange

- The historical currency ledger records historic currency exchange rates across 5 currency pairings, including several FIAT currencies and two cryptocurrencies (BTC and MEW).
- In the user dashboard our team created a chart which tracks the fluctuations in the currency value.
- Integrated into data is AI functionality, which helps with currency hedging. Essentially the large language model analyzes the historical data and offers hedging strategies based on the currency volatility.

4.3.2.5 Invoices

- Invoices are crucial for the financial module as they help keep track of the company's liquidity.
- The invoice chart is connected to the general ledger and helps the finance team categorize the incoming and outgoing invoices.

4.3.3 Physical ER Diagram:



4.3.4 Data Model and Key Entities:

Entity	Description	Key Relationships
GENERAL_LEDGER	Records all financial transactions with debit/credit amounts, accounts, and accounting periods.	Linked to INVOICES via GL_ENTRY_ID, to COST_CENTERS, and to PAYMENTS
INVOICES	Tracks billing documents between interplanetary	Connected to GENERAL_LEDGER, PAYMENTS (via

	parties with payment status and tax details.	PAYMENT_ID), and vendor/customer data
COST_CENTERS	Manages organizational units with budgets and planetary locations.	Associated with GENERAL_LEDGER entries via COST_CENTER_ID
EXCHANGE_RATES	Stores currency conversion rates for interplanetary transactions.	Used by GENERAL_LEDGER and PAYMENTS for foreign currency transactions
PAYMENTS	Records payment transactions including method, status, and interplanetary processing details.	Linked to INVOICES and GENERAL_LEDGER

4.4 Human Resource (HR) Module

Module	Key Functions	Description
HR		
HR Director	Full HR System	Complete access: applicant management, resume review, performance tracking, departmental trials coordination, promotion decisions, system configuration
HR Officer	HR Operations	Applicant screening, interview scheduling, basic employee data, payroll processing, attendance tracking
Department Head	Department-Specific HR	View applicants assigned to their department, provide trial period feedback, performance ratings, promotion recommendations

Hiring Manager	Recruitment Focus	Review assigned applicants, conduct interviews, update application status, skills assessment
Employee	Self-Service Portal	View personal data, update contact information, submit leave requests, access payroll information

4.4.1 Module Purpose:

The Human Resources (HR) module enables the end-to-end management of recruitment, talent development, and employee lifecycle data to support strategic workforce planning. It acts as the central hub for attracting, evaluating, and nurturing talent by integrating applicant tracking, trial performance, promotion workflows, and AI-driven placement. This unified system allows HR teams to make data-informed decisions, optimize departmental fit, and foster long-term employee growth.

4.4.2 Core Functionalities:

4.4.2.1 Applicant Tracking System (ATS) - CRUD operations

- Create employee profiles, job postings, trial records.
- Read resume data to extract related previous experiences and significant skills from their resumes.
- Update applicant status, employee records, trial results.
- Delete outdated applicant records and expired job postings.

4.4.2.2 Recruitment Management

- Tracks which candidate applications require processing or follow-up.
- Schedules interviews within one month from the application date.
- Collects and displays candidate evaluations from relevant department heads.
- **AI Job Notice Generation**
    - Automatically generates tailored recruitment notices for each department.
    - Uses department name and required skill sets as inputs to a language model.
    - HR teams can modify the AI-generated content through editable prompts, reducing time spent on repetitive writing tasks.

- Ensures consistency, clarity, and relevance in job postings while saving manual effort.

4.4.2.3 Departmental Trial System

- Monitors performance ratings of applicants during department trials.
- **AI Trial Analyzer:**
  - Leverages language models to analyze applicant data during trial periods, including: Trial duration, Performance ratings, Pass/fail outcomes
  - Provides actionable insights into training effectiveness and departmental readiness.
  - Helps departments identify performance gaps and adjust training programs accordingly.
- **AI Job Matching:**
  - Recommends the most suitable departments for high-performing trial employees.
  - Bases recommendations on skill sets, previous experience, and trial performance.
  - Automatically generates explanation narratives for HR transparency and candidate clarity.
  - Helps define optimal career paths, improving talent retention and internal mobility.

4.4.2.4 Career Development

- **AI Trial Analyzer - Skill Set Evaluation**
  - Identifies which skill sets are statistically linked to success (e.g., high ratings, promotions).
  - Guides HR teams in:
    - Designing relevant training programs.
    - Prioritizing development areas that lead to advancement.

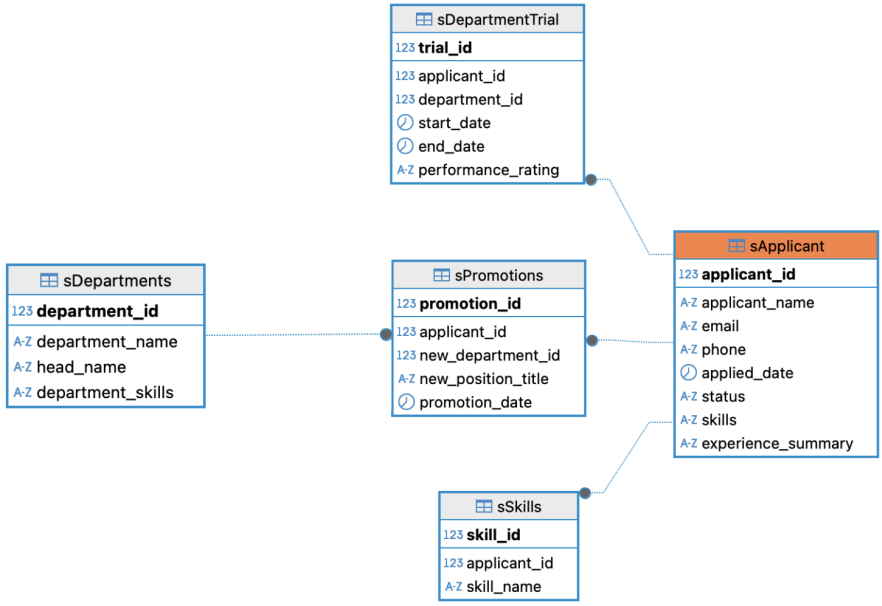
- Strengthens the company's talent pipeline and supports data-driven career development planning.

- Tracks employees selected for promotion after departmental trials.
- Maintains talent pipeline data to evaluate training and department effectiveness via trial pass/fail rates.

4.4.2.5 Employee Records

- Stores comprehensive employee data for long-term tracking and development.
- Supports strategic talent evaluation by maintaining historical performance and progression records.

4.4.3 Physical ER Diagram:



4.4.4 Data Model and Key Entities:

Entity	Description

Applicants	Represents individuals applying to the organization. Each applicant can have multiple skills, undergo several department trials, and receive multiple promotions. Central entity linking skills, trials, and promotions.
Skills	Defines specific competencies. Each skill can be associated with multiple applicants, establishing a many-to-many relationship between applicants and skills.
Departments	Defines organizational units. Each department has a name, a head, and a set of required skills. Departments are involved in multiple trials and promotions.
Department Trials	Records temporary assignments of applicants to departments for evaluation. Each trial involves one applicant and one department, applicants can only undergo one trial, and departments can host multiple applicants.
Promotions	Captures records of applicants being promoted. Each promotion includes the applicant, the new department, and the new position title. An applicant may receive only one promotion, and a department may be involved in many.

4.4.5 Entity Relationships

- An applicant can have multiple skills.
- A skill can be possessed by multiple applicants.
- An applicant can only undergo one department trial.
- A department trial is for one applicant and one department.
- An applicant can only receive one promotion.
- A promotion is associated with one applicant and a new department.
- A department can have multiple applicants in trials and multiple promotions.

5. Error Reporting

5.1 Purpose:

In a high-stakes environment like Mars, system reliability is mission-critical. The **Error Reporting** protocol ensures that bugs, glitches, data inconsistencies, or operational anomalies are swiftly documented, escalated, and resolved to maintain the integrity of resource management and human survival.

5.2 Reporting Channels:

Type	Contact Method	Details Required
System Bugs	techsupport@miacorp.mars	Module name, error description, time of occurrence, user role, screenshots (if available)
Data Discrepancy	dataaudit@miacorp.mars	Table/entity affected, suspected incorrect value, ID or record reference, impact assessment
Access Issues	accesscontrol@miacorp.mars	User ID, attempted action, access level, error message (if shown)
AI Decision Errors	ai.review@miacorp.mars	AI module used (e.g., Family Planning Recommendation), citizen ID involved, expected vs actual outcome

5.3 Emergency Response Protocol:

If the system impacts **life-critical functions** such as oxygen allocation or medical request processing:

- Immediately contact the **On-Duty Admin via secure channel SOS://admin.miacorp.mars.**
- Log the incident in the **Redline System Report** interface within the admin dashboard.
- AI failsafe will auto-route the case to the Critical Systems Team for manual override and investigation.

5.4 Log Capture and Versioning:

- **Error logs** are automatically recorded in the backend, tagged with timestamps, user IDs, and module names.
- Daily reports are aggregated and analyzed by the **Martian Ops QA Team**.
- System patches are version-controlled and deployed via the Starlink-optimized GitSync pipeline.

5.5 User Guidance:

A built-in "**Report Issue**" button is available on all system interfaces. Click to auto-fill current page context, browser info, and user session ID to reduce user effort in technical issue reporting.

“When Earth’s dust no longer holds our roots, may wisdom—not conquest—guide our steps across the red soil.“

For in the silence of Mars, ***Sage Orun*** sees not escape, but the echo of who we choose to become.

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